

We are writing as members of the UK Exoplanet research community to express our great concern regarding the proposed 30% cuts to the Science and Technology Facilities Council's (STFC) particle physics, astronomy, and nuclear physics (PPAN) programme. The UK Exoplanet community has leading roles in virtually all major international exoplanet projects and cuts to PPAN funding threaten the vitality of the exoplanet community as well as its leadership on the world stage.

The UK Exoplanet research community holds an annual meeting, which this year was held in Bristol. With a record number of attendees, the meeting illustrated just how this community has grown since the first UK Exoplanet Community Meeting held in Cambridge in 2014. It is an incredibly vibrant and active community that is dominated by excellent early career researchers who are playing key roles in driving exoplanet research in the UK. As such, we are particularly concerned by the potential for the proposed PPAN cuts to cause long-term damage to the standing of UK science.

Notwithstanding the proposed cuts, this is an extremely exciting time for exoplanet research. We are moving from an era of detecting exoplanets to one of understanding their formation, determining their internal structure, probing the composition of their atmospheres, and characterising their climates. UK researchers have led numerous extremely successful ground-based projects as well as playing leading roles in exoplanet science with the James Webb Space Telescope (JWST). The launch of the European Space Agency's (ESA) PLATO mission in 2027, first-light of the Extremely Large Telescope (ELT) later this decade, and the launch of the UK-led Ariel mission in the early 2030s also indicate a very bright future. The UK exoplanet community is also preparing for a major role in the Habitable Worlds Observatory (HWO), a space observatory with an expected launch date in the 2040s.

One of the next great scientific frontiers will be the search for life on planets outside the Solar System and the UK Exoplanet community is primed to play a key role in this endeavour. Not only does this align strongly with the STFC's science challenges in frontier physics, the UK Space Frontiers Astronomy and Astrophysics panel ranked "life on other planets" as the top science priority, and ranked HWO as the top mission priority. It is also a topic that is of great interest to the broader public. In a study looking at how learners in different cultures relate to science and technology, Sjøberg & Schreiner (2010) found that the possibility of life outside Earth is the one research question that unified people from all backgrounds, nationalities and genders. Essentially, everyone wants to know if there is life outside Earth.

However, like other areas of astronomy, particle physics, and nuclear physics (PPAN), the excitement generated by our research is overshadowed by the planned cuts to STFC's PPAN programme. The current intention is to cut PPAN funding to something like 70% of the funding level in 2024/2025. This level of cut will do enormous damage to exoplanet research and will have a particularly adverse effect on early career researchers, many of whom may lose their jobs or prospects for ones in the future. Since the UK Exoplanet community is dominated by very active early career researchers, the proposed cuts would severely hamper the UK's capacity to contribute to what is very obviously a topical and high impact research area.

According to the STFC Astronomy Grants Panel (AGP) Chair's report in 2026, funding for postdoctoral researchers has already dropped since 2024. While we welcome the recent assurance that postdoctoral numbers will be protected from further cuts, we note that these are to be fixed at the 2025/26 level that already include very significant cuts to an historically low level. We remain seriously concerned that this assurance locks us in to a level that leaves UK competitiveness in astronomy unsustainable. We should also stress that it's not just concern about funding for postdoctoral researchers, funding for projects is also under threat. The STFC prioritisation exercise has highlighted 48 projects that are at risk of losing funding. This includes projects explicitly focussed on exoplanet science, but also includes others that provide support for exoplanet research. The proposed cuts risk making it extremely difficult for UK researchers to continue leading the ground-breaking research that these projects enable.

We understand that UK Research and Innovation (UKRI) is changing the way in which it allocates funding, but have yet to see a clear justification why this necessitates such large cuts to the PPAN programme, especially given a record funding award to UKRI. If overall funding is increasing and curiosity-driven research is protected, why are we seeing a reduction in funding for curiosity-driven research in PPAN areas? We have also seen little indication that any alternative options have been seriously investigated.

It is clearly challenging to manage a budget that provides facilities for research across numerous research councils, covers international subscriptions, and also directly funds research in specific areas (PPAN). However, any cuts to extremely successful research areas should be based on a careful assessment of the strategic priorities, and on community consultations, and must not simply be an unintended consequence of complex budget allocations. It is crucial for there to be more clarity about what is driving these planned cuts.

As should be clear, cuts of this magnitude will do immense damage to PPAN research. Given that some of the most exciting advances in exoplanet science are likely to happen within the next couple of decades, if these cuts do go ahead it will risk the UK dropping out of its leadership position and losing the chance to make major contributions to one of the most exciting scientific advances of this generation.

Yours sincerely,

On behalf of the undersigned researchers in the UK Exoplanet community.

Signatories:

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